

Trajectory Optimisation for a Steerable Airdrop in Unknown Wind

Guiding packages to a target for relief distribution, with no wind sensor on board.

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80,000

PEOPLE STILL UNREACHED, ASSAM FLOODS, JULY 2026

A cloudburst on 19 July 2026 flooded four districts of Upper Assam. **705,000 people were affected**, roads were gone, and air was the only way in.

So supplies were pushed out of helicopters. A payload released without guidance is carried by the air mass for the whole way down — it does not aim, it drifts. **In this model it lands 23.9 m off.** In a flood, 23.9 m is the wrong side of the water.



SCAN 1 OF 2 · START HERE

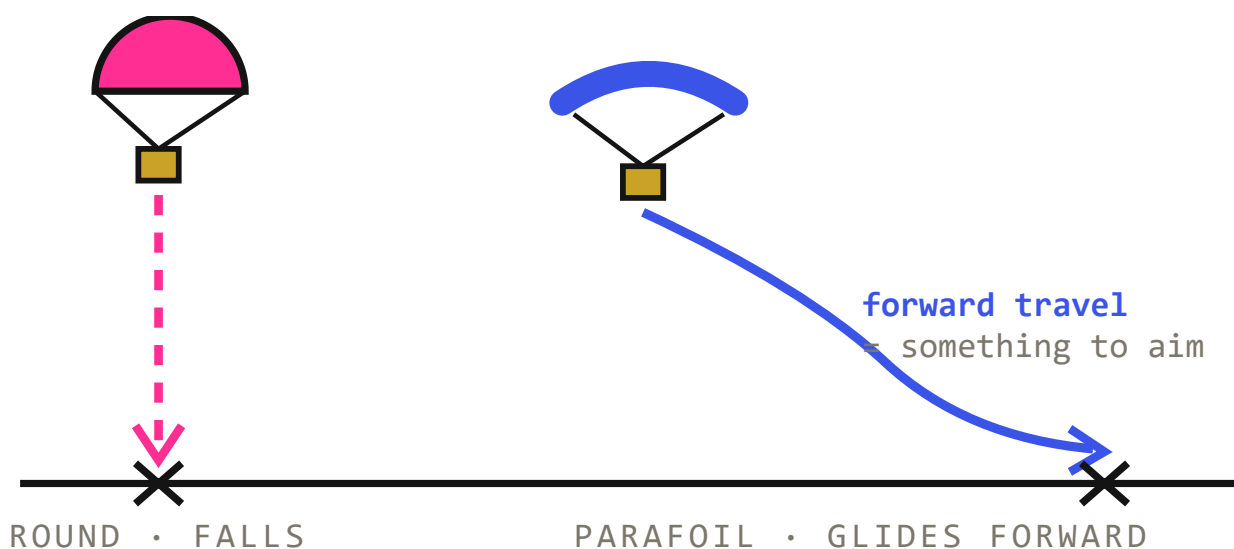
Watch it miss.

The drop above, falling in your own room. Point your phone at the floor and watch where it ends up.

ANSWER: YES, IF IT IS A WING

You can steer it. You cannot stop it.

A round parachute only falls. A **parafoil** is a ram-air canopy that **glides forward** as it descends, so pulling one side turns it. That forward travel is the only thing there is to aim.



$$v = \sqrt{(2mg / \rho S C_L)}$$

How fast it flies. Only the wing loading m/S appears — mass and canopy area never act separately.

But it has no engine and no brakes. It sinks at a steady rate, so **the moment it leaves the helicopter its time in the air is fixed** — and with it, the total distance it will ever cover.

That distance is almost always *more* than the distance to the target. It cannot slow down and it cannot circle forever. **It has to get rid of the surplus.**

WHAT IT WILL FLY BEFORE IT LANDS



DISTANCE TO THE TARGET



SO → It cannot fly straight at the target — it would arrive with height to spare.

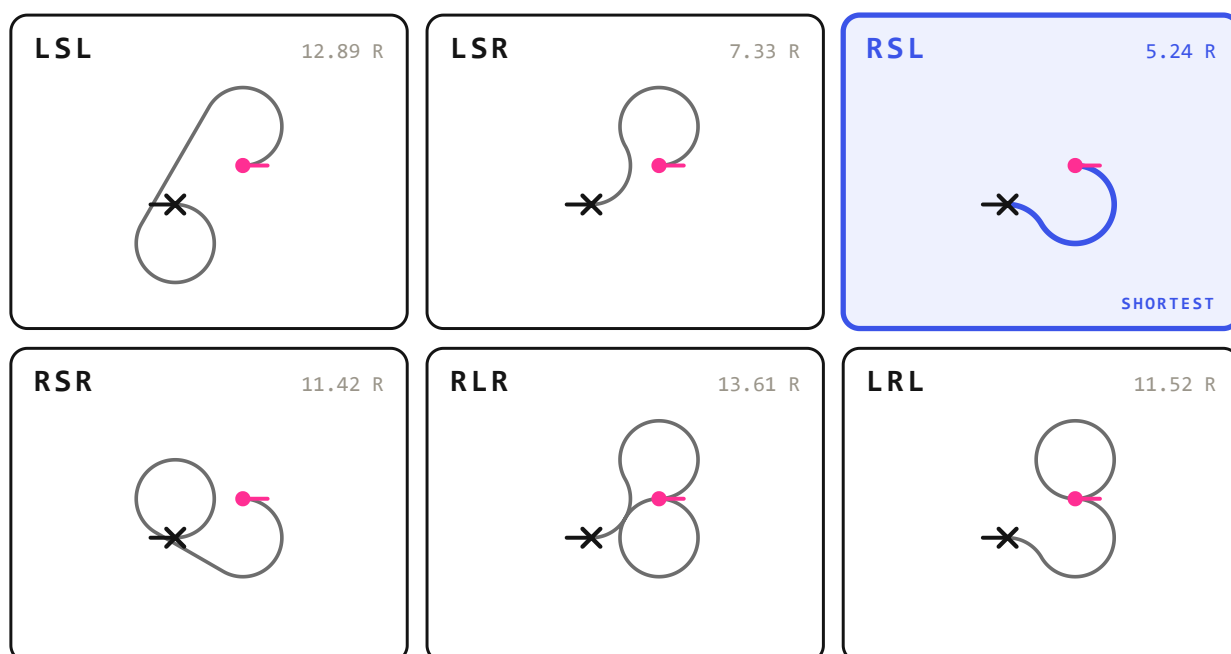
So it weaves — on purpose.

The guidance does not fly the shortest route. It flies a route of **exactly the right length**, holding a track angle α off the bearing to stretch the remaining path by $1/\cos \alpha$ until the budget is spent.

$$\cos \alpha = d / s$$

d = distance still to go · s = flying left in the budget. As height runs out, $d/s \rightarrow 1$ and $\alpha \rightarrow 0$ — a straight final approach, with no change of mode.

But d is not the straight line. It cannot turn on the spot, so the real distance to the target is a curve with a minimum radius. In **1957 Lester Dubins proved** the shortest such curve is always one of just six shapes.



Same start (dot) and target (cross), same heading at each end, solved six ways at one shared scale — computed by the model, not drawn. All six closed forms agree with step-wise integration to 9.0×10^{-12} m over 3,514 solutions.

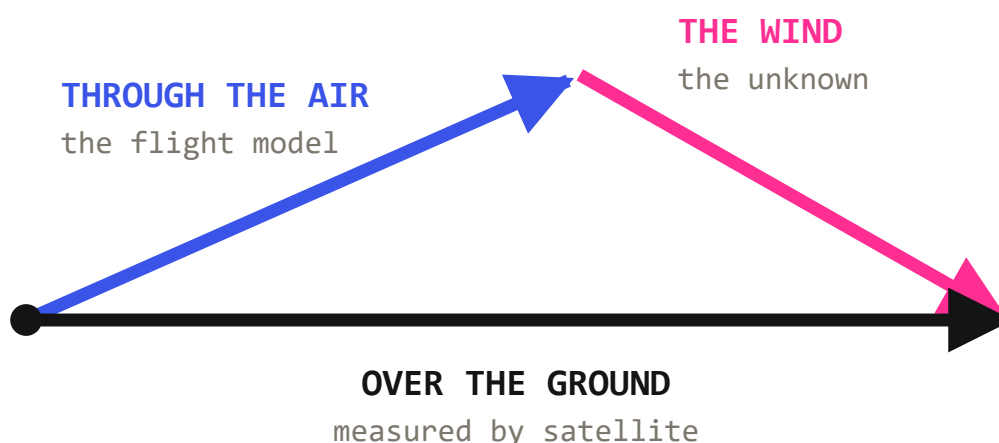
So it checks all six, takes the shortest, and measures its budget against *that*. **Dubins measures; the weave spends.**

S0 → From release to landing, it is almost always turning.

THE PART THAT PAYS FOR ITSELF

And that is how it finds the wind.

Nothing on board measures the wind and nothing is sent from the ground. An **Extended Kalman Filter** works it out from satellite velocity alone — because the wind is a subtraction:



State carried: $[w_x, w_y, v_a, b]$ — wind east and north, airspeed, and magnetometer bias. The bias enters through the heading trigonometry, which is what makes the problem non-linear.

Except it should be impossible

On a **fixed heading**, a headwind and an airspeed that is simply too low produce *exactly the same* track. No amount of data separates them — the state is unobservable. Flying straight, this cannot work at all.

FLYING STRAIGHT

Headwind, or an airspeed that is too low? **Same track.** Unsolvable.

TURNING

The two predict **different tracks**. The filter can tell them apart.

And by panel 03 it is *already turning*, for reasons that had nothing whatever to do with the wind. The manoeuvre that makes the wind measurable is one it was committed to anyway. **The measurement is free.**

0.104 m s⁻¹

MEDIAN WIND ERROR RECOVERED · 95% CI 0.096–0.112

BUT → Does any of it actually land closer?

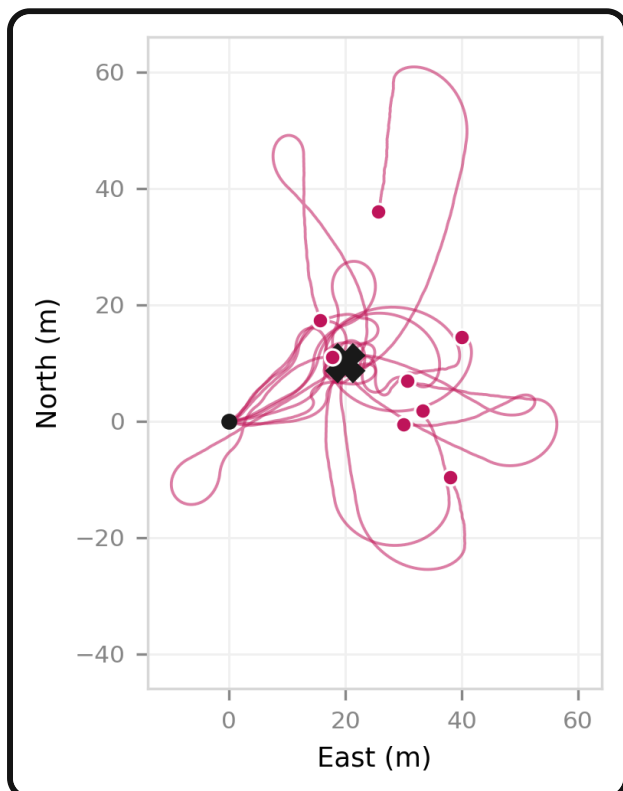
150 RANDOMISED DROPS

Yes — less than half the miss.

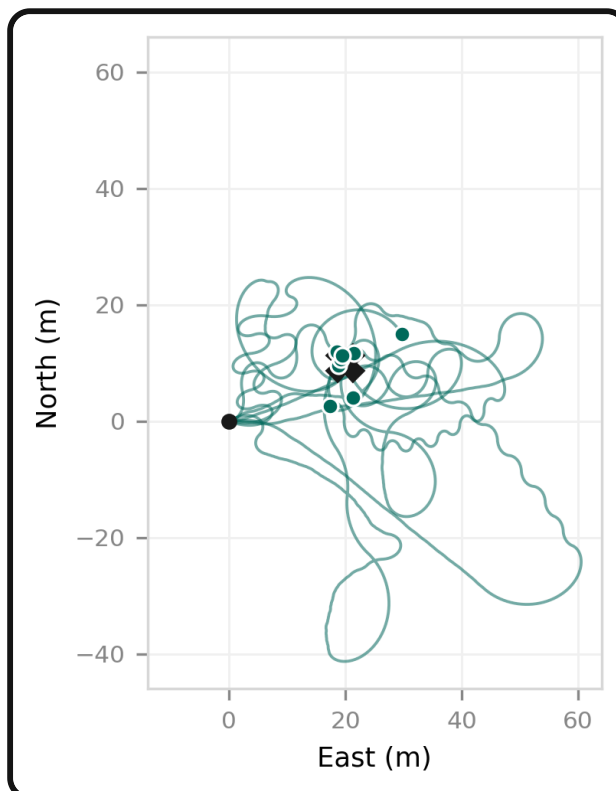
Identical conditions across rows. **Only the wind information available to the guidance differs.**

CONFIGURATION	MEDIAN MISS	90TH PCT
No estimate	8.57 m (6.49–12.46)	24.27 m
EKF onboard	3.89 m (2.85–5.28)	10.75 m
Told the true wind	4.62 m (3.37–5.44)	10.93 m

Intervals are 95% bootstrap percentile intervals on the median.



No estimate – touchdowns scattered.



EKF – touchdowns clustered on target.

Compared drop by drop, the EKF cuts the miss by a median **4.26 m** (95% CI 3.08–6.57) and wins in **112 of 150**.

What this does not show. The EKF and true-wind intervals overlap across most of their range, so those two are *not* separated at this sample size. Working the wind out on board is not shown to beat being handed it — only to beat not knowing it.

SO → Build the most capable canopy you can afford?

THE ONE NUMBER THAT DECIDES EVERYTHING

No. A heavier one is worse.

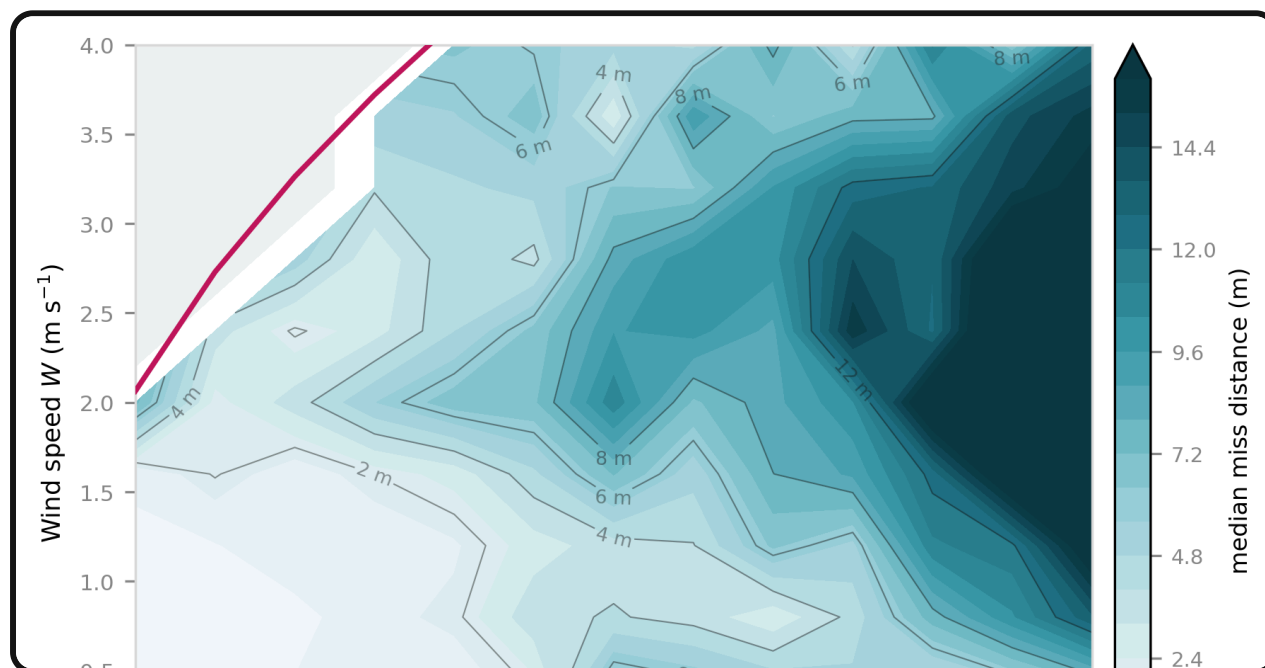
To make headway against wind of speed W it must fly faster than W , which forces wing loading up as W^2 . But turn radius rises *linearly* with loading — and turn radius is the smallest correction it can still make in the final seconds.

$$R_{\min} \propto W^2$$

The two requirements fight each other. The error floor follows the wind you *designed for*, not the wind you actually meet.

So building for a gale you never encounter makes you less accurate on a calm day. The right choice is the lowest loading that still penetrates, with margin — not the most capable one.

And loading is set by **ballast**. Unlike canopy area, it can be matched to the forecast on the morning of the drop.



Every combination of wing loading and wind speed, 24 drops each. The dark valley is the right answer, and it moves as the wind does.



SCAN 2 OF 2 · FINISH HERE

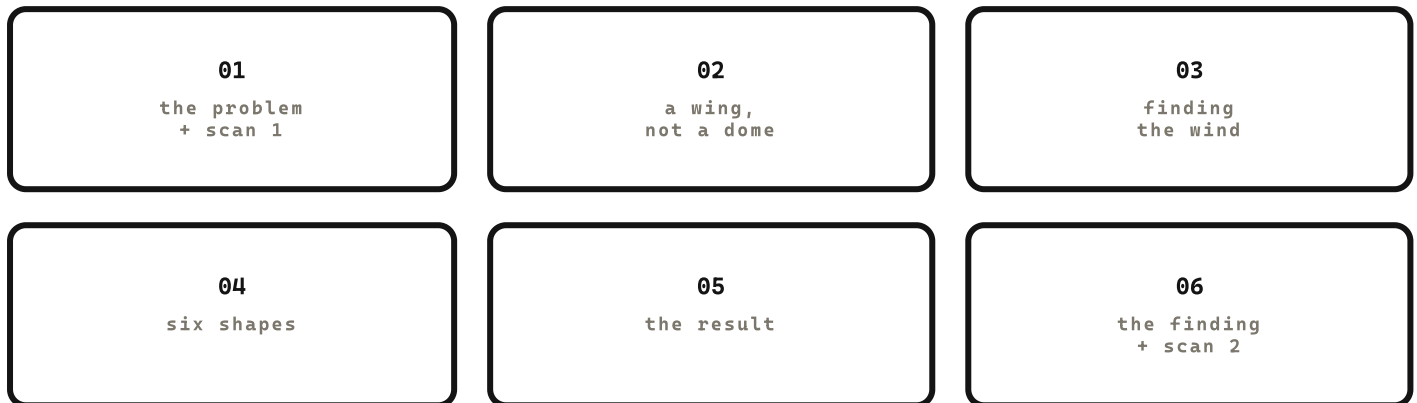
Now watch it land.

The same drop as scan 1 — same wind, same seed — flown with everything on this poster. The dotted line is where it went before.

✓ **23.9 m off, down to 2.35 m — at your feet.**

Sticking it together

Six sheets, three across and two down. Each panel is complete in itself, so **nothing needs trimming and nothing crosses a join**. The white border of each sheet becomes the gutter between panels.



1. Print pages 1–6 on A4 at **“Actual size” / 100%**, so the panels come out the size the layout was built for and the margins match. (The codes themselves were tested down to 70% and still scan — it is the layout that wants full size, not them.)
2. Lay all six out on the backing sheet *before* gluing anything. Line up the tops of panels 01–03, then 04–06.
3. Glue the middle sheet of the top row first and work outwards, so any error is shared between both sides instead of accumulating at one edge.
4. **Scan both QR codes with your own phone once the poster is up.** Test under the lighting in the hall, not at home.

If a QR code will not scan

Both codes were decoded from this very PDF at every resolution from draft to 300 dpi, so the artwork is not the problem. In practice it is nearly always **glare** — a reflection off tape, glue or a glossy sheet, sitting across the middle of the code. Move so the light is behind you rather than behind the phone. Failing that, low ink leaves gaps in the modules: reprint that one sheet. The codes carry the highest error correction level, so a crease or a fingerprint will not stop them.

Reading order

01 sets up the problem and sends them to the first scan. 02–04 are the model: why it can steer, how it finds the wind, how far it really has to fly. 05 is the evidence, including what it does *not* show. 06 is the finding, and sends them to the second scan — so a judge who only reads the top row still gets the problem, the mechanism and a demonstration.

Full paper and the interactive model: **1101-hub.github.io/guided-airdrop**